EDCA Prioritisation under 3GPP XR Traffic

A3: Literature review and revised project proposal

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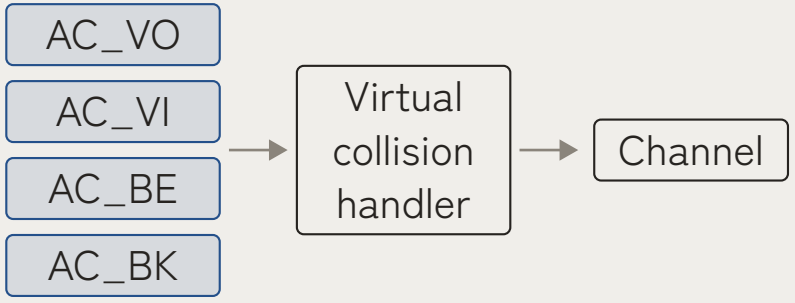
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1 Context

Where the project stands

1. **A1:** research question, basic path with the paper of Mangold et al. (2003)
2. **A2:** ns-3.48 set up, 3GPP XR traffic generator built and verified
3. **A3:** papers turned into a concrete experiment, proposal revised
4. **A4:** reproduce the baseline figure in ns-3
5. **Final:** replace the paper's traffic with a 3GPP model

EDCA in one slide



One backoff entity per access category (AC)

$$\mathrm{AIFS}[\mathrm{AC}] = \mathrm{SIFS} + \mathrm{AIFSN}[\mathrm{AC}] \cdot \mathrm{aSlotTime}$$

	AC_VO	AC_VI	AC_BE	AC_BK
AIFSN	2	2	3	7
CWmin	3	7	15	15
CWmax	7	15	1023	1023
TXOP limit	1.504 ms	3.008 ms	0	0

Smaller AIFSN and CW: higher priority.
Same slot: higher AC transmits, lower AC backs off.
Mangold et al.: one frame per TXOP for all ACs.

The research question

- Mangold et al., Fig. 8: under rising load, EDCA restrains the lower ACs
- Their traffic: 512 byte packets, exponential inter-arrival times, chosen deliberately simple
- The authors note that correlated arrivals of realistic traffic would affect the results
- XR video is periodic and bursty: about 10 kB per eye buffer, fragmented into several packets

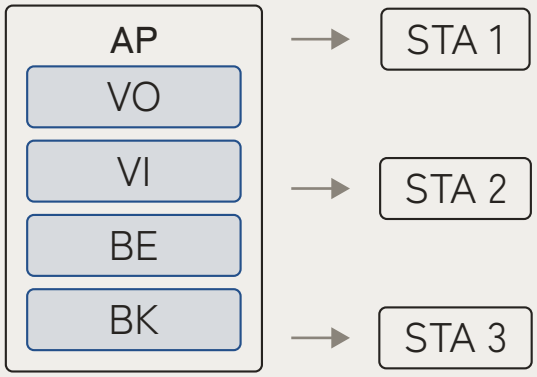
> **Does the prioritisation still hold when AC_VI carries bursty 3GPP traffic at the same mean load?**

2 Literature

Three papers, three roles

Paper	Type	Role in the project
Mangold et al., IEEE Wireless Commun., 2003	Protocol analysis, simulation	Baseline: Fig. 8
Kong et al., IEEE JSAC, 2004	Analytical model (3D Markov chain), ns-2	Effects of AIFS and CW, starvation
Huang and Liao, IEEE Trans. Wireless Commun., 2007	Analytical model (2D Markov chain, queueing), ns-2	802.11a timing, delay and jitter definitions

Mangold et al.: the baseline



12 downlink flows: one per AC to each station

- Only the AP sends data: contention only between its four ACs
- Poisson traffic, 512 byte, offered load per AC swept from 0 to 30 Mb/s
- 802.11a, all frames at 24 Mb/s, one frame per TXOP, no RTS/CTS
- Result: AC_BK, AC_BE and AC_VI drop out in turn, AC_VO grows until saturation
- Not reported: simulator, simulation time, seeds, retry limit, MSDU lifetime

Two analytical EDCA models

Kong et al. 2004

- One Markov chain per AC: backoff stage, counter, remaining frozen time
- Models AIFS freezing, the EDCA countdown rule, retry limit
- Finding: AIFS differentiates more strongly than CW, low priority ACs can starve

Huang and Liao 2007

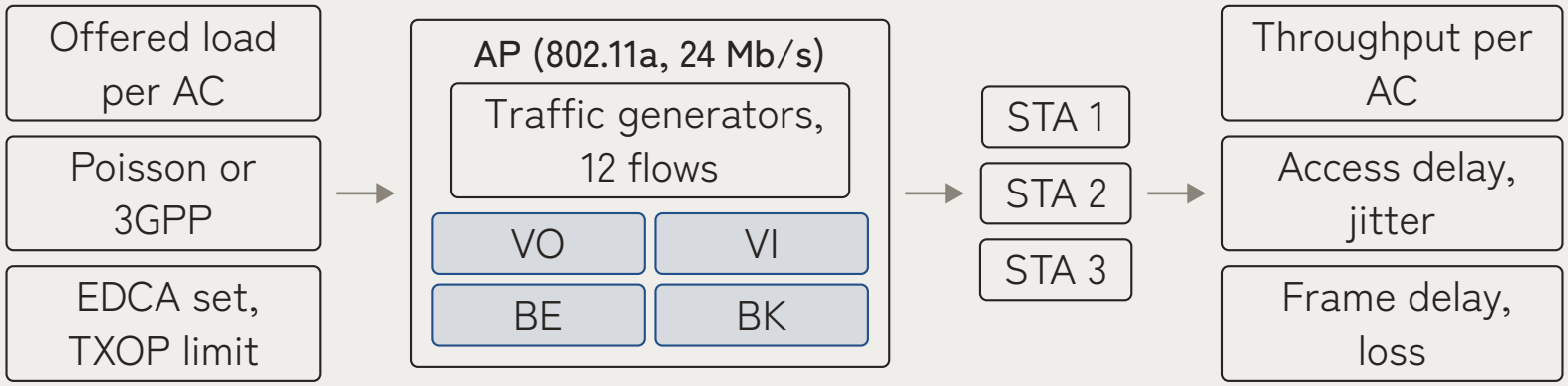
- Two dimensional chain per AC, plus the probability that the counter decrements
- Separates virtual and external collisions
- Adds access delay, jitter and queueing delay
- Finding: delay and jitter of low priority ACs grow steeply with the number of stations

What this means for the experiment

1. Fig. 8 has one transmitter: both models reduce to an **analytical cross-check** for the saturated end.
2. Kong et al. is not reproducible in ns-3 (SIFS of 20 µs, PHY header only given in bits). Huang and Liao is closest to Mangold et al. (802.11a, 24 Mb/s, four ACs).
3. **Delay and jitter** definitions taken from Huang and Liao.
4. All three: one frame per TXOP and simple traffic. **Bursty traffic with TXOP bursting** is the open gap.

3 Revised proposal

System architecture



Reference: Fig. 8 of Mangold et al. and an analytical EDCA model with one station

Three experiment scenarios

	Scenario	Purpose	When
1	Fig. 8 in ns-3: Poisson, Table 1 parameters, one frame per TXOP	Reproduce the order of the drop of the lower ACs	A4
2	Analytical model with one station and ns-3 frame durations	Independent check of the saturated end	A4
3	3GPP stream on AC_VI at the same mean load, TXOP limit 0 and ns-3 default	Effect of burstiness and of TXOP bursting	Final

What changed since A1

A1	A3	Why
250 kb/s per stream	Load sweep, 0 to 30 Mb/s per AC	x axis of Fig. 8; 250 kb/s belongs to Fig. 9
Control frames at 6 Mb/s	All frames at 24 Mb/s	Stated for the Fig. 8 section
Collisions not specified	Only virtual collisions in the AP	Single transmitter
Different simulators: open	Analytical cross-check	Kong et al., Huang and Liao
Throughput, delay, loss	Plus access delay, jitter, loss by cause	Definitions of Huang and Liao
Rate scaling: open	Sweep the 3GPP mean rate on the Fig. 8 axis	Equal mean load, same PHY
One frame per TXOP	TXOP limit as a parameter of scenario 3	All papers assume one frame per TXOP

XR traffic: eye buffer timing

- TR 26.926 allows both: eye buffers sent at the same time, or staggered
- Recommended configurations (Table 6.4-1): 7 of 8 send at the same time
- frame size M per eye buffer stays the same
- A2 used staggered: change to same time

$$M = \frac{R \cdot 10^6}{2F \cdot 8}$$

Timing	Burst
Same time	$2M$ every $1/F$
Staggered	M every $1/(2F)$

4 Next steps

A4: baseline reproduction

1. Verify the cause of the A2 delays in Wireshark
2. Repeat the traffic verification on the extended generator (Poisson and 3GPP)
3. Switch the eye buffer timing to “same time”
4. Check for saturation: first with increased capacity, then monitor queue drops in every run
5. Implement scenarios 1 and 2

- Same mean load, different burstiness: does EDCA still protect the lower ACs?



Thank you

Questions?